



Fig. 4 Symptoms of disease caused by *Pythium myriotylum* on cannabis plants with and without application of two biocontrol agents. (A) Lalstop was applied 7 days before inoculation with *Pythium*. Disease was rated after 18 days of incubation at 30°C. The plant on the left is the pathogen only treatment, the plant on the right received Lalstop. (B) Rootshield was applied 7 days before inoculation with *Pythium*. The root system shown on the left received the pathogen only while the one on the right received Rootshield. Roots were removed from the pots 18 days after inoculation with *Pythium*. Results from experiment 1 are shown.

against *Pythium*, and had no significant effect on root weights in either *Pythium* experiment.

Some of the differences in the efficacy of biocontrol treatments at managing disease caused by *P. myriotylum* between experiments may be attributed to the amounts of pathogen inoculum and biocontrol products used in each experiment. It is possible that the level of inoculum used in the first trial was insufficient to cause adequate disease; therefore, the effects of the biocontrol agents could not be distinguished from the control. Conversely, the amount of biocontrol product applied in the first experiment may not have been adequate to have actually managed disease in any significant way. More research is needed to determine to what degree the rate of application of biocontrol products has an effect on their efficacy at managing both *Fusarium* and *Pythium* on cannabis.

On cannabis cuttings, the inoculum causing initial infection may originate from within the environment in which the cuttings are rooted, which is kept warm (24–27°C) and humid (70–90% relative humidity). Spores of *F. oxysporum* have been detected in the air in the rooting rooms, as well as in recirculating water (Punja 2020b). In addition, cuttings may have incipient infections originating from the stock plants, where *F. oxysporum* was shown to be transmitted internally and asymptotically,

likely present in the pith and xylem tissues (Punja 2020b). Therefore, biocontrol agents applied to cuttings should be able to colonize the cut surface, the emerging roots, as well as grow internally in the pith tissues. In this study, the biocontrol agents were applied 48-hr prior to pathogen inoculation to ensure this colonization had occurred. Endophytic colonization, defined as recovery following surface-sterilization of tissues, was observed to occur by both *Gliocladium* and *Trichoderma* spp. in the 0–10 cm stem segments of cuttings, suggesting that this could be one of the mechanisms through which disease was reduced, i.e. through competitive exclusion of the pathogen.

The active ingredient in Lalstop (previously known as Prestop) is *G. catenulatum* (*Clonostachys rosea* f. *catenulata*) strain J1446. In previous research, this biocontrol agent was shown to significantly reduce root and stem rot severity and seedling mortality on greenhouse cucumbers, caused by *F. oxysporum* f. sp. *radicis-cucumerinum*, when it was applied as a drench 24-hr or 48-hr prior to inoculation with the pathogen (Rose et al. 2003). In addition, applications of *G. catenulatum* on spinach plants controlled *F. oxysporum* f. sp. *spinaciae* wilt, as well as *Fusarium* wilt caused by *F. foetens* on Hiemlais begonia (Tian and Zheng 2013). Preventative